

OOP LAB 10

Problem Statement

You are tasked to design a Mobile Phone Management System in C++ using inheritance.

MobilePhone.h

```
class MobilePhone {
public:
    MobilePhone();
    MobilePhone(const string& manufacturer, const string& model);
    ~MobilePhone();

    void turnOn();
    void turnOff();
    bool isPoweredOn() const;

    std::string getManufacturer() const;
    std::string getModel() const;
    void setManufacturer(const string& manufacturer);
    void setModel(const string& model);

    void showInfo() const;

private:
    string manufacturer;
    string model;
    bool poweredOn;
};
```

ApplePhone.h

```
class ApplePhone : public MobilePhone {  
public:  
    ApplePhone(const string& model, const string manufacturer);  
  
    void makeVoiceCall(string& phoneNumber);  
    void endVoiceCall();  
    void sendSMS(const string& message, long recipientNumber);  
    void takePhoto(const string& filePath);  
    void openBrowser(const string& url);  
};
```

AndroidPhone.h

```
class AndroidPhone : public MobilePhone {  
public:  
    AndroidPhone(const string& model, const string manufacturer);  
  
    void makeVoiceCall(const string& phoneNumber);  
    void endVoiceCall();  
    void sendSMS(const string& message, long recipientNumber);  
    void takePhoto(const string& filePath);  
    void openBrowser(const string& url);  
};
```

WindowsPhone.h

```
class WindowsPhone : public MobilePhone {
public:
    WindowsPhone(const string& model, const string manufacturer);

    void makeVoiceCall(const string& phoneNumber);
    void endVoiceCall();
    void sendSMS(const string& message, long recipientNumber);
    void takePhoto(const string& filePath);
    void openBrowser(const string& url);
};
```

Example Output

ApplePhone calling 03001234567...

Call ended.

Sending SMS to 03001234567: Hello!

Photo saved to C:/Photos/Apple/apple_img1.jpg

Browser opened at www.apple.com

AndroidPhone calling 03007654321...

Call ended.

Sending SMS to 03007654321: Hi!

Photo saved to D:/Photos/Android/android_img1.jpg

Browser opened at www.android.com